

Habitat for Humanity

560 8th Street

Castlegar, BC

GENERAL NOTES:

- ALL WORK TO CONFORM TO THE BRITISH COLUMBIA BUILDING CODE LATEST EDITION, LOCAL CODES AND BY-LAWS OF AUTHORITIES HAVING JURISDICTION.
- ALL WORK TO BE PERFORMED WITH RESPECT TO GOOD BUILDING PRACTICES.
- CONTRACTOR TO CAREFULLY INSPECT THE SITE OF WORK AND BE FULLY INFORMED OF EXISTING CONDITIONS AND LIMITATIONS
- NO WORK TO COMMENCE WITHOUT PROPER PERMITS AND LICENSES.
- MEASUREMENTS, GRADES AND LEVELS ARE TO BE VERIFIED AT THE SITE BEFORE CONSTRUCTION.
- CONTRACTOR TO CHECK AND VERIFY ALL DIMENSIONS, ELEVATIONS, DRAWINGS, DETAILS AND SPECIFICATIONS AND REPORT ALL ERRORS OR DISCREPANCIES TO THE OWNER PRIOR TO PROCEEDING WITH THE WORK.
- CONTRACTOR TO VERIFY LOCATIONS AND DETAILS OF ALL CONCEALED SERVICES. PROTECT AND RELOCATE WHERE INDICATED ALL SERVICES FROM DAMAGE DURING CONSTRUCTION PERIOD..
- CONTRACTOR TO MAKE GOOD AND REPAIR ALL EXISTING PARTS AND SURFACES DAMAGED BY DEMOLITION OR NEW CONSTRUCTION, REFRESH TO MATCH SURROUNDING AREA BETWEEN CORNERS OR ABUTMENTS COMPLETE.
- DEMOLISH WHERE NOTED, AND REMOVE DEBRIS FROM SITE, MINIMIZE DISRUPTION TO NEIGHBOURS. ALL SALVAGE MATERIAL (TO BE CONFIRMED BY OWNER) REMAIN THE PROPERTY OF THE OWNER UNLESS OTHERWISE NOTED.
- VERIFY LOCATION OF ALL UNDERGROUND LINES WITHIN THE AREA OF CONSTRUCTION PRIOR TO COMMENCING EXCAVATION. NOTIFY OWNER AT TIME OF EXCAVATION.
- DETERMINE LOCATION OF PARTITIONS NOT DIMENSIONED BY THEIR RELATION TO COLUMN FACE OR CENTRE, WINDOW JAMB OR MULLION, OR OTHER SIMILAR FIXED ITEM.
- DO NOT DRILL OR CUT FLOOR JOISTS, BEAMS, COLUMNS OR OTHER STRUCTURAL ELEMENTS UNLESS SPECIFICALLY INDICATED. DRILL SLABS WHERE APPROVED. CORE DRILL CIRCULAR OPENINGS THROUGH SLABS. LINE DRILL OR SAW CUT RECTANGULAR OPENINGS.
- PROVIDE BLOCKING FOR SOLID BACKING BEHIND ALL WALL AND CEILING MOUNTED DOOR HARDWARE, ACCESSORIES, MILLWORK, PLY EDGES, MISC. METAL ITEMS, GYPSUM BOARD EDGES ETC.
- TAPE, FILL AND SAND ALL NEW G.W.B.
- INSTALL CARBON MONOXIDE DETECTORS TO SATISFY BCBC 2018 (9.32.4.2 'CARBON MONOXIDE ALARMS')
- ALL FLASHING TO BE PREFINISHED TO SUIT OWNERS COLOUR SCHEME. FLASHING TO BE INSTALLED AT ALL CHANGES IN HORIZONTAL EXTERIOR FINISHES AND OVER ALL UNPROTECTED EXTERIOR OPENINGS. CAULKING TO BE INSTALLED AROUND ALL UNFLASHED EXTERIOR OPENINGS. FLASHING TO BE INSTALLED AT ALL PENETRATIONS IN THE ROOF SYSTEM AND AT ALL CHANGES IN THE ROOF PLANE.
- VAPOUR BARRIER TO MIN. 6 MIL. SEAL ALL JOINTS AND HOLES TO PREVENT LEAKAGE. PROVIDE ALSO 12" WIDE LAPS BELOW SLAB ON GRADE.
- A FREE VENT AREA OF 1/300 OF THE INSULATED ATTIC AREA SHALL BE PROVIDED AT THE ROOF, APPROXIMATELY HALF FROM THE EAVES AND HALF FROM THE TOP. (WITH NOT LESS THAN 25% OF THE OPENINGS AT THE TOP OF THE SPACE & NOT LESS THAN 25% OF THE OPENINGS AT THE BOTTOM OF THE SPACE. SEE BCBC 2018 9.19 ROOF SPACES)
- PROVIDE GASKET TO U/S OF SILL PLATES. (POLYETHYLENE FILM OR TYPE S ROLL ROOFING)
- DESIGN AND FINISHING INTENT TO REACH MINIMUM STEP 3 OF BC ENERGY STEP CODE

FIELD REVIEWS:

- WSA ENGINEERING (2012) LTD. (WSA) PROVIDES FIELD REVIEW FOR THE WORK SHOWN ON THE STRUCTURAL DRAWINGS PREPARED BY WSA. THIS REVIEW IS A PERIODIC REVIEW AT THE PROFESSIONAL JUDGEMENT OF WSA. THE PURPOSE IS TO ASCERTAIN THAT THE WORK IS IN GENERAL CONFORMANCE WITH THE PLANS AND SUPPORTING DOCUMENTS PREPARED BY WSA AND TO FULFILL THE REQUIREMENTS FOR THE COMPLETION OF LETTERS OF ASSURANCE REQUIRED BY THE APPLICABLE BUILDING CODE.
- THE CONTRACTOR IS RESPONSIBLE TO UNDERTAKE THE WORK IN ACCORDANCE WITH THE DESIGN DRAWINGS, THE BCBC AND GOOD CONSTRUCTION PRACTICES. INSPECTIONS BY WSA WHETHER UNDERTAKEN OR NOT FOR ANY ITEM DO NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY TO COMPLETE THE WORK AS DIRECTED BY THE CONTRACT DOCUMENTS OR DRAWINGS, BEST PRACTICES AND THE BCBC. THE CONTRACTOR WILL SAVE HARMLESS AND INDEMNIFY WSA, ITS OWNERS OR EMPLOYEES FROM ANY DAMAGES RESULTING FROM POOR WORKMANSHIP, ERRORS OR OMISSIONS BY THE CONTRACTOR.
- ALL NON-CONFORMING WORKS THAT REQUIRE REMEDIAL ACTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR, ANY EXTRA TIME OR COST INCURRED TO WSA, TO ASSIST OR ADVISE THE CONTRACTOR IN RECTIFYING THE WORK SHALL BE BORNE BY THE CONTRACTOR.
- ENSURE THAT WORK TO BE INSPECTED IS COMPLETE AT THE TIME OF INSPECTION AND IN ACCORDANCE WITH THE CONTRACT DOCUMENTS. ADDITIONAL INSPECTIONS REQUIRED DUE TO INCOMPLETE WORK OR POORLY EXECUTED WORK, AS JUDGED BY WSA AS WELL AS ADDITIONAL DESIGN OR REMEDIAL WORK CAUSED BY DEVIATIONS FROM THESE DRAWINGS, MAY BE CHARGED TO THE GENERAL CONTRACTOR AT THE DISCRETION OF WSA.
- A MINIMUM OF 48 HOURS NOTICE SHALL BE GIVEN BY THE CONTRACTOR FOR ANY INSPECTION TO BE CARRIED OUT BY WSA. INSPECTIONS ARE REQUIRED PRIOR TO CONCEALING ANY STRUCTURAL WORK SHOWN ON THESE DRAWINGS.

MINIMUM REQUIRED FIELD REVIEWS FOR THIS PROJECT:

A. FOOTINGS:

INSPECT WHEN REBAR AND FORMS ARE INSTALLED

B. FOUNDATION WALLS:

INSPECT WHEN REBAR IS INSTALLED AND STILL VISIBLE PRIOR TO BEING FULLY COVERED BY FORMS

C. FRAMING:

INSPECT ALL STRUCTURAL ELEMENTS DETAILED IN THESE DRAWINGS PRIOR TO BEING COVERED WITH INSULATION, SIDING, OR OTHER ELEMENTS

CONCRETE:

- PROVIDE CONCRETE AND PERFORM WORK TO CSA-A23.3.
 - MINIMUM 28 DAY COMPRESSIVE STRENGTHS AS INDICATED BELOW. ALL CONCRETE NORMAL WEIGHT - 150 PCF, TYPE 10 CEMENT, TYPE F FLYASH, MAXIMUM 3/4" AGGREGATE FOR ALL CONCRETE EXCEPT 1 1/4" MAXIMUM AGGREGATE FOR CHUTE PLACED SLABS ON GRADE. SUBMIT PROPOSED MIX DESIGN TO THE ENGINEER FOR APPROVAL.
- | LOCATIONS | STRENGTH MPa (PSI) | AIR % | SLUMP +20mm | EXPOS. CLASS |
|-------------------------|------------------------|------------|-------------|--------------|
| FOOTINGS | 25 (3600) | 1-4 | 70 | - |
| SUSPENDED SLABS & BEAMS | 25 (3600) | 4-7 | 70 | F2 |
| RETAINING WALL | 25 (3600) | 4-7 | 70 | F2 |
| INTERIOR S.O.G. | 25 (3600) | 1-4 | 60 | - |
| EXPOSED S.O.G. | 32 (4640) | 4-8 | 60 | C2 |
| WALLS & COLUMNS | 25 (3600)
30 (4350) | 1-4
4-7 | 70
70 | -
F2 |
- DO NOT USE ADMIXTURES OTHER THAN AIR ENTRAINMENT, STANDARD WATER REDUCERS OR SUPER PLASTICIZERS WITHOUT PRIOR APPROVAL OF THE ENGINEER.
 - REJECT ALL CONCRETE WHEN TIME BETWEEN BATCHING AND PLACING EXCEEDS 2 HOURS.
 - DO NOT ADD WATER TO THE CONCRETE ON SITE UNLESS AUTHORIZED BY THE ENGINEER.
 - CONSOLIDATE ALL CONCRETE USING MECHANICAL VIBRATORS.
 - CONTROL JOINTS FOR SLAB-ON-GRADE: SAWCUT TO A DEPTH OF 25% OF SLAB THICKNESS AS SOON AS POSSIBLE AND NO LATER THAN 20 HOURS AFTER POURING AT MAXIMUM 6.1m SPACING OR AT LOCATIONS SHOWN ON THE DRAWINGS.
 - CONSTRUCTION JOINTS: AS SHOWN ON THE DRAWINGS OR AS DIRECTED BY THE ENGINEER.
 - PROTECT CONCRETE FROM ADVERSE WEATHER CONDITIONS IN ACCORDANCE WITH CSA A23.1, A23.3
 - CONSTRUCT FORMWORK IN ACCORDANCE WITH WCB REGULATIONS AND CSA S269.3. FORMWORK DESIGN IS THE RESPONSIBILITY OF THE CONTRACTOR.

REINFORCING:

- NEW DEFORMED BARS TO CSA G30.18 GRADE 400 (60 KSI). WELDED WIRE FABRIC TO CSA G30.5. ANCHOR BOLTS TO ASTM A307.
- PLACE REINFORCING BARS TO CSA A23.1. TIE ALL BARS SECURELY IN PLACE TO PREVENT DISPLACEMENT. SUPPORT SLAB REINFORCING ON SUITABLE CHAIRS OR SUPPORTS AT MAXIMUM 4 FT. CENTRES. PROVIDE CORNER BARS TO MATCH HORIZONTAL WALL REBAR.
- PROVIDE CLEAR CONCRETE COVER FOR REBAR AS FOLLOWS:

SURFACE POURED AGAINST GROUND	3"
FORMED SURFACE EXPOSED TO GROUND OR WEATHER	2"
BEAMS	2" TO MAIN STEEL
COLUMNS	2" TO MAIN STEEL
WALLS	1 1/2"
SLABS ON GRADE	1 1/2"
- SPLICE REBAR AS FOLLOWS (UNLESS OTHERWISE NOTED):

BAR SIZE-	25M	20M	30M	15M	10M
LAP SPLICE-	51"	31"	71"	25"	18"
- MINIMUM 2-15M REINFORCING AROUND OPENING LARGER THAN 12" AT EACH SIDE OF OPENING. EXTEND 2'-0" PAST CORNER.
- CONTRACTOR TO PROVIDE 48 HOURS NOTICE FOR REBAR INSPECTION.
- WHERE SUSPENDED SLAB DRAWINGS ONLY SHOW PRINCIPAL REINFORCING IN ONE DIRECTION, PROVIDE SHRINKAGE AND TEMPERATURE REINFORCING PERPENDICULAR TO PRINCIPAL REINFORCING AND LOCATE BETWEEN MAIN TOP AND BOTTOM REINFORCING, PER PLANS.
- PROVIDE CORNER BARS FOR ALL HORIZONTAL WALL REINFORCING
- PLACE REINFORCING BARS UNIFORMLY AND SYMMETRICALLY, U.N.O.
- WHERE NEW CONCRETE POUR MEETS ABUTTING CONCRETE, DRILL AND GROUT ALL LONGITUDINAL REINFORCING & I.N.O. DRILLING AND GROUTING OF REINFORCING SHALL BE WITH 'HILTI' HY-150 SYSTEM OR APPROVED EQUAL
- NO WELDING OF ANY CONCRETE REINFORCING STEEL IS PERMITTED WITHOUT WRITTEN APPROVAL FROM THE STRUCTURAL ENGINEER.

WOOD CONSTRUCTION:

- ROUGH CARPENTRY SHALL BE CONSTRUCTED IN ACCORDANCE TO THE HIGHEST INDUSTRY STANDARDS AND TO THE REQUIREMENTS OF PART9 AND PART 4 OF BCBC 2018.
- SILL PLATES TO BE PRESSURE TREATED, LEVELLED AND FASTENED TO FOUNDATION WALL WITH 1/2" Ø ANCHOR BOLTS (UNLESS NOTED OTHERWISE (UNO)) EMBEDDED MIN. 4" @ 8'-0" o/c. MAX. (OR IF SHEAR WALL AS PER DETAIL) WITH MIN. 2 IN EACH SILL.
- ALL TRUSSES TO ENGINEERED AND INSTALLED TO MANUFACTURERS SPECS. PROVIDE ALL GIRDERS, HANGERS, SUPPORTS, HARDWARE, BRACING, ETC. AS REQUIRED. MANUFACTURER TO BRING TO THE ATTENTION OF OWNER/CONTRACTOR ANY FURTHER BEARING REQUIRED FOR TRUSSES PROVIDED.
- TRUSS/JOIST MANUFACTURER TO PROVIDE ALL PERTINENT DRAWINGS AND DESIGN INFORMATION INCLUDING MEMBER REACTIONS TO STRUCTURAL ENGINEER FOR REVIEW PRIOR TO CONSTRUCTION.
- ALL BEARING COLUMNS OF GIRDER TRUSSES TO AND SUPPORT BEAMS ARE TO BE POSTED TO FOUNDATION.
- ALL LINTELS TO EXTERIOR OR BEARING WALLS TO BE (2) - 2x10 UNO.
- BUILT-UP SAWN LUMBER BEAMS AND POSTS SHALL BE CONNECTED WITH 3.5" COMMON NAILS IN TWO ROWS AT 10" O/C IN EACH FACE.
- HEADER JOISTS EMBEDDED IN CONCRETE TO BE TREATED.
- FRAMING HANGERS, ANCHORS, AND CLIPS SHALL BE PRE-ENGINEERED GALVANIZED METAL FABRICATION TO SUIT THE LOADING AND SPAN OF THE FRAMING MEMBERS SUPPORTED. ALL SPECIFIED HARDWARE IS AS MANUFACTURED BY SIMPSON STRONGTIE. ALTERNATIVE MUST BE APPROVED BY ENGINEER.
- PROVIDE JOIST HANGERS AT FLUSH FRAMED WOOD MEMBERS.
- DOUBLE OR TRIPLE STUD UNDER LINTELS AND BEAMS, AS REQUIRED OR UNO.
- WHEN BLOCKING OR BRIDGING IS REQUIRED BY BCBC 9.23.9.4 SPACING SHALL BE NO GREATER THAN 6'10" O/C (SEE 9.23.9.4 BCBC 2018).
- PROVIDE SOLID BLOCKING BETWEEN JOISTS OVER BEARING WALLS AND BEAMS. TOE NAIL THE BLOCKING TO THE WALL PLATE WITH 3" COMMON NAILS AT 8" O/C STAGGER SIDE TO SIDE AS MIN OR AS NOTED FOR WALL PLATE NAILING IN THE WALL SCHEDULE. NAIL THE FLOOR OR ROOF SHEATHING TO THE BLOCKING. NAIL WALL PLATES ABOVE TO THE BLOCKING.
- SOLID BLOCKING TO BE INSTALLED FOR ADEQUATE SUPPORT OF TOWEL BARS, CURTAIN AND CLOSET RODS, SHELVES, GRAB BARS AND SIMILAR FIXTURES WHERE REQUIRED.
- MULTI-PLY LVL'S SHALL BE CONNECTED AND INSTALLED AS PER MANUFACTURER'S SPECIFICATIONS.
- GLULAM BEAMS TO BE PRE-DRILLED TO ACCEPT BOLTS, SHEAR PLATES, ETC.
- ALL BOLTS, NUTS, AND WASHERS IN CONTACT WITH GLULAM TO BE HOT DIPPED GALVANIZED. ALL EXPOSED BOLT HEADS TO BE RECESSED INTO GLULAM MEMBERS.
- ORIENTATE FLOOR AND ROOF SHEATHING WITH FACE GRAIN PERPENDICULAR TO JOISTS. STAGGER PANEL JOINTS.
- ROOF AND FLOOR SHEATHING TO BE FASTENED WITH 2.5" COMMON NAILS AT 6" O/C PANEL EDGES AND BLOCKING. 12" O/C SPACING AT INTERMEDIATE SUPPORTS OR UNO.
- FLOOR SHEATHING TO BE GLUED DOWN TO SUPPORT PRIOR TO FASTENING.
- WALL SHEATHING TO BE FASTENED AS PER WALL SCHEDULE. NAILING SPACING TO BE REDUCED BY 1/3 WHERE AUTOMATIC NAILS ARE USED. DO NOT OVERDRIVE NAILS. PROVIDE DOUBLE STUDS AT PANELS JOINTS IN WALLS WHERE NAIL SPACING IS 2" OR LESS.
- STAIR FRAMING IN ACCORDANCE WITH PART 9 BCBC 2018.
- GRADE AND SPECIES OF FRAMING AS FOLLOWS: (UNO ON DRAWING)
 - SOLID SAWN BEAMS, POSTS, COLUMNS, HEADERS, LEDGERS, & JOISTS, TO BE NO. 1 OR BETTER, DOUGLAS FIR/ LARCH
 - STUDS, BUILT-UP BEAMS, BUILT-UP POSTS, JOISTS, BUILT-UP HEADERS, & LEDGERS, TO BE NO. 1/2 OR BETTER SPF
 - EXTERIOR WALL SHEATHING TO BE 1/2" O.S.B. OR 1/2" PLYWOOD
 - ROOF SHEATHING TO BE MIN. 1/2" DF PLYWOOD UNLESS OTHERWISE NOTED.
 - ALL SUBFLOORING TO BE MIN. 3/4" T&G PLYWOOD UNLESS OTHERWISE NOTED.

NON-STRUCTURAL COMPONENTS:

- NON-STRUCTURAL COMPONENTS ARE NOT THE RESPONSIBILITY OF WSA ENGINEERING LTD. BUT ARE DESIGNED, DETAILED, SPECIFIED AND REVIEWED IN THE FIELD BY OTHERS. LETTERS OF CERTIFICATION OF ADEQUACY, INSTALLATION, ETC, OF SUCH COMPONENTS ARE BY OTHERS.
- MANUFACTURERS OF NON-STRUCTURAL COMPONENTS WHICH AFFECT THE STRUCTURAL FRAMING SHALL SUBMIT SHOP DRAWINGS TO THE ARCHITECT AND WSA ENGINEERING LTD. FOR REVIEW. THE SHOP DRAWINGS SHALL CLEARLY INDICATE THE LOAD IMPOSED ON THE STRUCTURE. REVIEW WILL BE LIMITED TO THE EFFECT OF THE COMPONENTS ON THE STUCTURAL FRAMING.
- EXAMPLES OF NON-STRUCTURAL COMPONENTS INCLUDE BUT ARE NOT LIMITED TO:
 - ARCHITECTURAL COMPONENTS SUCH AS HANDRAILS, GUARDRAILS, RAILINGS, FLAG POST, REMOVABLE CANOPIES, CEILINGS, VEHICLE PROTECTION SYSTEMS, ORNAMENTAL COMPONENTS
 - ARCHITECTURAL PRECAST CONCRETE AND ITS ATTACHMENTS
 - ARCHITECTURAL GLASS BLOCKS AND THEIR ATTACHMENTS
 - BRICK AND BLOCK VANEERS, REINFORCING, AND TIES
 - LANDSCAPING COMPONENTS SUCH AS BENCHES, LIGHT POSTS, PLANTERS
 - CURTAIN WALL SYSTEMS, CLADDING, SKYLIGHT, WINDOW MULLIONS
 - INTERIOR AND EXTERIOR NON-LOADING STEEL STUD WALLS
 - SUPPORT AND BRACINGS OF MECHANICAL AND ELECTRICAL SYSTEMS AND EQUIPMENT FOR NON-GRAVITY AND SEISMIC LOADS
 - WINDOW WASHING EQUIPMENT AND ITS ATTACHMENTS
 - ELEVATORS, ESCALATORS, AND OTHER CONVEYING SYSTEMS, INCLUDING PROPRIETARY SUPPORT BEAMS AND ATTACHMENTS
 - NON-STRUCTURAL MASONARY

DESIGN LOADS (Castlegar) PER BCBC 2018

- SPECIFIED DEAD LOADS:

ROOF	15 PSF	(0.72 kPa)
FLOOR	15 PSF	(0.72 kPa)
- SPECIFIED LIVE LOADS:

FLOOR	40 PSF	(1.915 kPa)
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- CLIMATIC DATA:

GROUND SNOW (Sg)	87.74 PSF (4.2 kPa)
ROOF SNOW (S)	50.34 PSF (2.41 kPa)
RAIN (Sr)	2.09 PSF (0.1 kPa)

WIND LOADS:

(1/10)	5.64 PSF	(0.27 kPa)
(1/50)	7.1 PSF	(0.34 kPa)

SEISMIC LOADS:

Sa(0.2) = 0.129
Sa(0.5) = 0.1
Sa(1.0) = 0.074
PGA = 0.058

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ENGINEERING (2

Civil Structural

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